Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/25/2006
PI Name:	Sukharev, Sergei Ph.D.		
Project Title:	Mechanosensitive Channels In Bacteria: The Role Of Protein-Lipid Interactions In The Mechanism Of Gating		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Task Description:	We are currently approaching only a minimal level of understanding of how cells sense and respond to mechanical stimuli. There still remains a drastic mismatch between the amount of 'functional' phenomenology data, experimental observations collected at the organismal and cellular levels, and corresponding molecular descriptions of the underlying mechanosensory mechanisms. MscL and MscS are bacterial channels which respond to membrane stretch by opening large aqueous pores. They reside in the inner membrane of Escherichia coli and serve as protective safety valves in the event of osmotic shock. Solved crystal structures tremendously facilitate progress toward elucidating the molecular mechanisms of these tension-driven valves. They also aid in the search for a general understanding of the biophysical principles that govern these primary events in mechanosensation. During the previous funding periods, we determined the overall structure of MscL, and evaluated the energetic and spatial parameters that characterize its conformational transitions. We have built a plausible molecular model of MscL conformational transition. Since MscS was identified		

	and crystallized later, there is currently less data on its mechanisms and the models of its gating are less precise. Our methodology includes site-directed mutagenesis, covalent protein modifications, patch-clamp measurements, thermodynamic and kinetic analysis of dose-response curves and transition rates, video-microscopy, molecular modeling, and simulations. The goal is a consistent biophysical description of gating of these two structurally dissimilar mechanosensitive channels at an atomic-scale level.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research leads to understanding of very basic mechanisms of force detection by macromolecules in living organisms. It will help to clarify general principles and mechanisms of mechanosensation, which encompass a wide range of phenomena from simple osmoregulation in bacteria to complex phenomena such as gravitropism in plants and balance and hearing in humans. MscL represents simple and powerful model system highly useful for molecular and biophysical experiments as well as for molecular computation. Obtaining a complete and self-consistent picture of mechanosensitive channel gating which includes the protein, surrounding lipids and solvent would be a stepping stone to understanding the mechanisms of the growing families of sensory channels modulated by membrane stretch cell volume, lipids, amphipathic substances, or temperature.
Task Progress:	During the past year we experimented on several biophysical aspects of the two bacterial mechanosensitive channels: MscL and MscS. We have established the critical role of pore hydration in the energetics of gating transitions in the Mechanosensitive Channel of Large conductance MscL. Analysis of mutants with polar or charged substitutions in the main hydrophobic gate of the channel provided valuable information about gating intermediates. We have accomplished thermodynamic analysis of several such mutants to clarify the nature and position of low-conducting conformations (substates) in the transition pathway. This analysis showed that the closed-to-substate transition in WT MscL is associated with only a minor conductance increment, but the largest in-plane area expansion and free energy change. The V23D and G22N mutations, which make the main gate more ‘polar’, strongly affect this sub-transition by reducing the in-plane area changes and the energy difference associated with it. The measured area changes indicate that closed conformations of these mutants are physically pre-expanded. Our data support the two-gate mechanism in which the first sub-transition can be viewed as opening of the central (M1) gate, resulting in an expanded water-filled ‘leaky’ conformation. Strong facilitation of this step by polar substitutions suggests that the separation of M1 helices, associated with hydration of the pore in WT MscL, is the major energetic barrier for opening. These data also suggest that the bulk of the conductance gain in the second sub-transition occurs through the opening of the N-terminal (S1) gate. The recently solved crystal structure of the small conductance mechanosensitive channel (MscS) has been an invaluable tool in the search for the gating mechanism, however many functional aspects of the channel remained unsettled. To characterize the gating of MscS in E. coli spheroplasts we utilized a pressure clamp apparatus along with special software developed in-lab to generate and analyze dose-response curves. We found that MscS exhibits essentially voltage-independent activation by tension, but at the same time strong voltage-dependent inactivation under depolarizing conditions. The MscS activation curves obtained under saturating ramps of pressure, at different voltages, gave estimates for the energy, area, and gating charge for the closed-to-open transition as 24 kT, 18 nm2, and +0.8 respectively. Perhaps the most salient and intriguing property of MscS gating was a strong dependence on the rate of pressure application. MscS prefers to respond in full to abrupt stimuli but manages to ignore those applied slowly. We proposed a kinetic scheme and gating mechanism to account for the observed phenomenology in the framework of available structural information.
Bibliography Type:	Description: (Last Updated: 03/01/2007)
Abstracts for Journals and Proceedings	Akitake B, Anishkin A, Sukharev S. "The dashpot mechanism of stretch-dependent gating in MscS. " Presented at the 49th Annual Meeting of Biophysical Society, Long Beach, Calif, February 12-16, 2005. Biophys J. 2005; 88 (1):289A-289A Part 2 Suppl. S . , Jan-2005
Articles in Peer-reviewed Journals	Akitake B, Anishkin A, Sukharev S. "The “dashpot“ mechanism of stretch-dependent gating in MscS." J Gen Physiol. 2005 Feb;125(2):143-54. PMID: 15657299 , Feb-2005
Articles in Peer-reviewed Journals	Anishkin A, Chiang CS, Sukharev S. "Gain-of-function mutations reveal expanded intermediate states and a sequential action of two gates in MscL. " J Gen Physiol. 2005 Feb;125(2):155-70. PMID: 15684093 , Feb-2005
Articles in Peer-reviewed Journals	Anishkin A, Sukharev S. "Explicit channel conductance: Can it be computed? " Biophys J. 2005 Jun;88(6):3742-3. PMID: 15764668 , Jun-2005
Articles in Peer-reviewed Journals	Chiang CS, Shirinian L, Sukharev S. "Capping transmembrane helices of MscL with aromatic residues changes channel response to membrane stretch. " Biochemistry. 2005 Sep 20;44(37):12589-97. PMID: 16156671 , Sep-2005
Books/Book Chapters	Sukharev S, Anishkin A, Chiang C-S, Betanzos M, Guy HR . "MscL, a Bacterial Mechanosensitive Channel." in "Bacterial Ion Channels and Their Eukaryotic Homologs." Ed. A. Kubalski, B. Martinac. Washington DC : ASM Press. c2005, p.259-290., Jul-2005